

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1102

COURSE OUTCOME COVERED

CO5: Analyze object-oriented software using quality assurance and usability testing Techniques (BL4, BL5)
Assessment Tool Weightage: 25%

Total Marks:25

QUESTIONS: 05

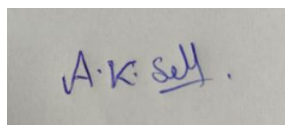
Annexure A

Error Analysis Task

Code of Conduct:

*I A K Selva Prabhu(192421224) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

A rectangular box containing a handwritten signature in blue ink that reads "A.K. Selva".

RUBRICS FOR CALCULATION:

Error Analysis Task

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Error Identification	20%	Accurately identifies all errors with clear understanding of issues	Identifies most errors with minor omissions	Identifies limited errors; some are missed	Fails to identify key errors
Root Cause Analysis	30%	Clearly explains underlying causes with strong technical reasoning	Explains causes with moderate clarity	Partial explanation; weak reasoning	Unable to identify causes of errors
Correction & Justification	25%	Provides correct solutions with strong justification and logic	Provides correct corrections with some justification	Corrections are partially correct or weakly justified	Incorrect or no corrections provided
Application of Concepts	15%	Effectively applies relevant concepts to analyze and resolve errors	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts
Clarity & Presentation	10%	Presents analysis clearly, logically, and systematically	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Annexure A

Error Analysis Task

Q1. Scenario: Online Banking System

Modules:

User Login, Fund Transfer, Account Verification

Question:

Write pseudocode for a quality assurance testing process that validates secure user authentication and fund transfer operations. Include input validation, transaction verification, exception handling, and test result logging.

Q2. Scenario: E-Commerce Shopping Platform

Features:

Product Search, Shopping Cart, Payment Gateway

Question:

Design pseudocode for a usability testing evaluation that measures customer interaction efficiency, including product search time, checkout completion time, and navigation complexity. Explain how the algorithm collects and analyzes user behavior data.

Q3. Scenario: Hospital Management System

Modules:

Patient Registration, Appointment Scheduling, Medical Records

Question:

Write pseudocode to generate and execute test cases for appointment scheduling functionality. Include validation of doctor availability, appointment conflicts, patient information accuracy, and system responses to invalid requests.

Q4. Scenario: Airline Reservation System

Modules:

Passenger Registration, Flight Booking, Seat Allocation

Question:

Develop pseudocode for a constraint-based testing approach where booking requests must satisfy conditions such as valid passenger information, seat availability, and payment confirmation. Show how invalid booking scenarios are detected and handled.

Q5. Scenario: Smart Library Management System

Features:

Book Search, Book Reservation, Borrow/Return Services

Question:

Write pseudocode for integrating quality assurance and usability testing where the system evaluates both functional correctness (reservation and borrowing operations) and user experience factors such as response time, ease of navigation, and accessibility.

Q1. Online Banking System – Secure Authentication & Fund Transfer

Pseudocode:

START

Initialize Test_Log

// Test User Login

INPUT username, password

IF username is empty OR password is empty THEN

 Log "Invalid input"

ELSE

 Validate username and password

 IF credentials are correct THEN

 Perform Multi-Factor Authentication

 IF MFA is successful THEN

 Log "Login successful"

 ELSE

 Log "MFA failed"

 ENDIF

 ELSE

 Log "Invalid login credentials"

 ENDIF

ENDIF

// Test Fund Transfer

INPUT sender_account, receiver_account, amount

IF any input is empty OR amount <= 0 THEN

 Log "Invalid transfer input"

ELSE

Verify sender account
Verify receiver account
Check account balance

IF accounts are valid AND balance $\geq$ amount THEN
 Execute fund transfer

 IF transaction is successful THEN
 Verify transaction ID and updated balance
 Log "Fund transfer successful"
 ELSE
 Log "Fund transfer failed"
 Rollback transaction
 ENDIF

ELSE
 Log "Transaction rejected"
ENDIF
ENDIF

Store all test results in Test_Log

END

Key QA checks: Input validation, authentication security, transaction verification, exception handling, and result logging.

Q2. E-Commerce Platform – Usability Testing Pseudocode:

START

Initialize User_Test_Data

FOR each test user

 Record Start_Time

 // Product Search

 Ask user to search for a product

 Record Search_End_Time

 Search_Time = Search_End_Time - Start_Time

 Store Search_Time

 // Navigation

 Count number of clicks/pages used

 Store Navigation_Count

```

// Shopping Cart and Checkout
Add product to cart
Record Checkout_Start_Time

Complete checkout and payment
Record Checkout_End_Time

Checkout_Time = Checkout_End_Time - Checkout_Start_Time
Store Checkout_Time

IF checkout is successful THEN
    Record "Completed"
ELSE
    Record "Failed"
ENDIF

END FOR

Calculate:
    Average Search Time
    Average Checkout Time
    Average Navigation Count
    Checkout Success Rate

IF Search Time is low AND
    Checkout Time is low AND
    Navigation Count is low THEN
    Report "High Usability"
ELSE
    Report "Usability Improvements Required"
ENDIF

Generate User Behavior Report

END

How data is collected and analyzed:
The algorithm records search time, checkout time, number of clicks, navigation steps, and completion status for each user. The collected data is averaged and compared with predefined usability targets to identify slow operations and navigation difficulties.

```

Q3. Hospital Management System – Appointment Scheduling

Pseudocode:

START

Create Test_Cases

```
Test_Cases =  
    Valid Appointment  
    Invalid Patient Information  
    Unavailable Doctor  
    Appointment Conflict  
    Invalid Date/Time  
    Missing Required Information
```

```
FOR each Test_Case
```

```
    Execute Test_Case
```

```
    INPUT patient_id, doctor_id, date, time
```

```
    // Validate Patient
```

```
    IF patient information is invalid THEN
```

```
        Log "Patient validation failed"
```

```
        Continue to next test
```

```
    ENDIF
```

```
    // Validate Date and Time
```

```
    IF date or time is invalid THEN
```

```
        Log "Invalid appointment date/time"
```

```
        Continue to next test
```

```
    ENDIF
```

```
    // Check Doctor Availability
```

```
    IF doctor is unavailable THEN
```

```
        Log "Doctor unavailable"
```

```
        Continue to next test
```

```
    ENDIF
```

```
    // Check Appointment Conflict
```

```
    IF doctor already has appointment at same time THEN
```

```
        Log "Appointment conflict detected"
```

```
        Continue to next test
```

```
    ENDIF
```

```
    Create appointment
```

```
    IF appointment created successfully THEN
```

```
        Log "Appointment scheduling successful"
```

```
    ELSE
```

```
        Log "System response error"
```

```
    ENDIF
```


END FOR

Generate Test Result Report

END

Main validations: Patient accuracy, doctor availability, appointment conflicts, valid date/time, and proper handling of invalid requests.

Q4. Airline Reservation System – Constraint-Based Testing Pseudocode:

START

INPUT passenger_details

INPUT flight_details

INPUT seat_number

INPUT payment_details

// Constraint 1: Passenger Information

IF passenger_details are invalid THEN

 Log "Invalid passenger information"

 Reject booking

 STOP

ENDIF

// Constraint 2: Flight Validation

IF flight does not exist OR flight is unavailable THEN

 Log "Invalid or unavailable flight"

 Reject booking

 STOP

ENDIF

// Constraint 3: Seat Availability

IF requested seat is already booked THEN

 Log "Seat unavailable"

 Reject booking

 STOP

ENDIF

// Constraint 4: Payment Confirmation

Process payment

IF payment is not confirmed THEN

 Log "Payment failed"

 Reject booking

 STOP

ENDIF

```
// All constraints satisfied
Confirm flight booking
Allocate requested seat
Generate booking ID
```

```
Log "Booking successful"
```

```
END
```

Constraint-based approach:

The booking is accepted **only when all constraints are satisfied**. If even one condition fails, the booking is rejected and the reason is recorded in the test log.

Q5. Smart Library Management System – QA + Usability Testing

Pseudocode:

```
START
```

```
Initialize Test_Log
Initialize User_Feedback
```

```
// Functional Testing
Search for a book
```

```
IF book exists THEN
    Display book details
ELSE
    Display "Book not found"
ENDIF
```

```
Record Search_Response_Time
```

```
// Reservation Testing
Select available book
Request reservation
```

```
IF book is available THEN
    Create reservation
    Log "Reservation successful"
ELSE
    Log "Reservation failed"
ENDIF
```

```
// Borrow Testing
Request book borrowing
```

```
IF user is eligible AND book is available THEN
    Create borrowing record
```

```
    Log "Borrow successful"
ELSE
    Log "Borrowing rejected"
ENDIF

// Return Testing
Return borrowed book

IF borrowed book exists THEN
    Update book status
    Log "Return successful"
ELSE
    Log "Invalid return request"
ENDIF

// Usability Testing Measure:
Response_Time
Number_of_Clicks
Navigation_Steps
Accessibility_Errors

Collect User_Feedback

Calculate:
    Average Response Time
    Average Navigation Steps
    Accessibility Score
    Task Success Rate

IF functional tests pass AND
response time is acceptable AND
navigation is simple AND
accessibility is good THEN    Report
"System Quality and Usability Good"

ELSE

    Report "Improvements Required"

ENDIF

Save Test Results

END
```